

MTH209: Assignment 4

PCA Function

The goal in this assignment is to take a covariance matrix as input, and return the k -dimensional score for an input vector. Here, k and the input vector will also be inputs to the function. I explain more below:

The user will call your function by inputting a $d \times d$ covariance matrix, Σ and the goal is to return the first k principal component (PC) scores for an input vector x , where recall for a vector x , the PC scores are:

$$v_1^\top x, v_2^\top x, \dots, v_k^\top x.$$

You may assume that $k \leq d$ is a positive integer.

```
# covariance: d x d covariance matrix input
# x: a vector of length d lying in the d-dim space
# k: a positive integer (assumed less than equal to d)
PC_scores <- function(covariance, x, k)
{
  ...
  ...
  scores <- numeric(length = k)
  ...
  return(scores) # output must be a vector of length k
}
```

I will input your function and test the function for my own choice of the input covariance, k , and x .

INSTRUCTIONS: copy and paste ONLY the function in the `assignment4.R` file in the GitHub repository. NO PARTIAL MARKING for this assignment.